

Question:

Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .

Reasoning + Answer:

<think>

1. We need to find the sum of all integer bases $b > 9$ for which 17_b (which is 17 in base b) is a divisor of 97_b (which is 97 in base b).
2. To determine when 17_b is a divisor of 97_b , we first convert these numbers from base b to base 10 .
3. In base b , 17_b means $1 \cdot b + 7$. Converting 97_b to base 10 , we have $97_b = 9 \cdot b + 7$.
4. We need to check when $17_b = 1 \cdot b + 7$ divides $97_b = 9 \cdot b + 7$.
5. Therefore, we need to solve the congruence :
$$1 \cdot b + 7 \mid 9 \cdot b + 7$$
or equivalently ,
$$b + 7 \mid 9 \cdot b + 7.$$
6. We can rewrite this condition as :
$$b + 7 \mid 9b + 7 - b - 7.$$

Simplifying the right-hand side , we get :
$$b + 7 \mid 8b.$$
7. This implies that :
$$b + 7 \text{ is a divisor of } 8b.$$
8. Let $k = b + 7$. Then :
$$k \mid 8b \implies k \mid 8(b - 7).$$
9. This means :
$$k \mid 8b - 56.$$
10. Since $k = b + 7$, we have :
$$k \mid 8(b - 7) + 7k - k = 8k - 56.$$
11. Therefore, we need k to divide :
$$8k - 56.$$
12. For k to divide $8k - 56$, it must also divide 56 (since $8k$ will always be divisible by k).
13. The divisors of 56 are :
$$1, 2, 4, 7, 8, 14, 28, 56.$$
14. We need to check which of these values are greater than 9 :
15. Now, we find the corresponding values of b :
$$14 = b + 7 \implies b = 7 \text{ (not greater than 9)},$$

$$16 = b + 7 \implies b = 9 \text{ (not greater than 9)},$$

$$28 = b + 7 \implies b = 21,$$

$$56 = b + 7 \implies b = 49.$$
16. Thus, the valid bases b are 21 and 49 .
17. Finally, we sum these valid bases :
$$21 + 49 = 70.$$

< answer >

70

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